From firm-level Q&A change to sector ranking

1 CURRENT Q&A VS. OWN HISTORY

Current Q&A embedding $q_{i,t}$

Prior own-firm embedding mean $m_{i,t}$

$$d_{i,t} = q_{i,t} - m_{i,t}$$

Signed change from the firm's prior expanding history

2 WHY SIGNED CHANGE MATTERS

ANGULAR DISTANCE

$\theta(q, \bar{m})$

scalar angle only

SIGNED DISPLACEMENT

direction + magnitude

TWO SIGNED FORMS

RAW	UNIT
d	$u = d/\|d\|$

3 FIRMS → SECTOR RANKS (11 ETFS; 3 EXAMPLES)

Technology Health Care Industrials

Group firm-level measures by sector

Average one measure per firm within each sector

FIT EACH FORM SEPARATELY

RAW / UNIT · PCA-32 + Ridge

ANGULAR · native 1-D Ridge

4 RANKING & ALLOCATION

RAW-DISPLACEMENT RANKING

IC .158

mean monthly sector-rank IC

UNIT-DIRECTION TOP 1

NET RETURN

42.1% p.a.

SPY-ADJUSTED ALPHA

+27.0% p.a.